PROJECT REPORT

AI-Based Hyperlocal Demand Prediction for Dark Stores and Quick Commerce

Using Censored Demand Recovery and Ensemble Forecasting

Course: Artificial Intelligence and Machine Learning

Name	Roll No.
Assir Thota	23EC017
Ayush Vaibhav Gond	23EC020
Parth Sidhu	23EC040
Shreya Mohanty	23EC049
Srijan Gupta	23EC051

Submitted To

Dr. Anshika Srivastava

Assistant Professor, Program Coordinator (ECE)

Department of Electronics and Communication Engineering

Submitted in partial fulfillment of the requirements for the degree of

B.Tech in Electronics and Communication Engineering

(Specialisation: Rail Engineering)

SCHOOL OF TECHNOLOGY
GATI SHAKTI VISHWAVIDYALAYA

(A Central University under the Ministry of Railways, Govt. of India)

April 2026

Abstract

Accurate demand forecasting for perishable goods in dark stores and quick commerce is complicated by a fundamental data problem: stockout events artificially truncate observable sales, causing models trained on censored data to systematically underestimate true demand. This paper presents a two-stage pipeline that addresses this challenge using the FreshRetailNet-50K benchmark, the first open dataset providing verified hourly stockout annotations across 50,000 store-product time series from 898 dark stores in 18 cities. Stage 1 evaluates three lightweight gradient boosting and ensemble models — Random Forest, XGBoost, and LightGBM — for latent demand recovery, replacing the original paper's computationally intensive deep learning models. LightGBM with a Tweedie objective achieves the best Stage 1 result: 27.83% WAPE and +0.00% WPE after post-hoc bias calibration (calibration factor 1.0748), matching the paper's TimesNet benchmark of 27.62%, while XGBoost achieves comparable accuracy and Random Forest, lacking a Tweedie objective, shows degraded calibration. The demand-supply decoupling score $\rho_{DS} = +0.46$ indicates residual supply correlation attributable to the power-law demand distribution. Stage 2 trains three forecasting models on the recovered demand signal: a similar scenario average (SSA) statistical baseline, a per-horizon LightGBM, and a weighted ensemble of LightGBM and a tabular multilayer perceptron (MLP), all trained on the full 50,000 store-product pairs. The LightGBM achieves 29.36% mean WAPE and the ensemble 29.30%, both competitive with the paper's Temporal Fusion Transformer (29–31%), while the SSA baseline reaches 39.94%. Results confirm that the Tweedie objective is essential for accurate Stage 1 recovery, and that per-horizon calibration is the primary mechanism for controlling forecast bias. A six-module operational dashboard translates pipeline outputs into actionable inventory decisions.

Keywords: demand forecasting, censored demand, stockout recovery, dark stores, quick commerce, LightGBM, XGBoost, random forest, ensemble learning, perishable goods, MNAR

Contents

1 Introduction

The rapid growth of dark store-based quick commerce, characterized by hyperlocal 30-minute delivery from micro-fulfillment centers, has created acute demand forecasting challenges. Unlike traditional retail, dark stores operate with a narrow 3-kilometer service radius, a volatile customer base, and a product mix dominated by short-shelf-life perishables such as fresh produce, dairy, and seafood. These factors amplify demand uncertainty and stockout frequency.

A persistent yet underaddressed problem in this setting is *demand censoring*: when a product stocks out, the system records only the units sold before the stockout, not the true consumer demand. A store stocking out of fresh strawberries at 11:00 AM may record 100 units sold despite genuine demand for 200 units. If a forecasting model trains on this censored history, it systematically underestimates future demand, perpetuating stockouts in a self-reinforcing cycle.

Existing public retail forecasting benchmarks, including M5 [2], Corporación Favorita [3], and Walmart [4], do not provide stockout annotations, preventing distinction between true zero demand and supply-constrained zeros. The recently released FreshRetailNet-50K [1] addresses this gap by providing verified hourly stockout labels alongside hourly sales data, promotional discounts, and meteorological covariates.

This paper makes the following contributions:

- A two-stage demand estimation pipeline applied to FreshRetailNet-50K, with Stage 1 performing censored demand recovery and Stage 2 performing 7-day ahead forecasting on the recovered signal.
- A comparative evaluation of three lightweight Stage 1 models — Random Forest, XGBoost, and LightGBM — replacing the paper's deep learning recovery models, with analysis of the impact of the Tweedie objective on bias calibration.
- A LightGBM-based Stage 1 model that recovers latent demand to within 27.83% WAPE, reducing systematic bias from −7.37% to +0.00% via post-hoc scalar calibration, matching the paper's TimesNet benchmark of 27.62%.
- A full-data Stage 2 evaluation across three models — SSA (39.94%), LightGBM (29.36%), and a LightGBM+MLP ensemble (29.30%) — trained on all 50,000 store-product pairs, with both LightGBM and ensemble competitive against the paper's TFT (29–31%).
- Empirical evidence that per-horizon bias calibration (factors 1.05–1.11) is the primary control mechanism for WPE, and that the MLP's smooth Softplus activation yields the lowest systematic underestimation bias (−5.86% WPE) among Stage 2

models.

- An operational six-module web dashboard translating pipeline outputs into inventory decisions: demand overview, stockout countdown, lost sales audit, what-if causal simulation, smart replenishment, and weather impact analysis.

2 Terminology and Notation

Table 1 summarizes the key abbreviations, symbols, and technical terms used throughout this paper.

3 Related Work

3.1 Retail Demand Forecasting

Classical time series methods such as ARIMA [20] and exponential smoothing assume demand is accurately observed in sales data. Deep learning approaches including the Temporal Fusion Transformer [5], Informer [6], and Autoformer [7] have improved accuracy under normal conditions through attention mechanisms and covariate integration, but share the same flawed assumption: that sales equal demand. Tree-based gradient boosting, particularly LightGBM [10] and XGBoost [11], has emerged as a strong baseline for tabular retail forecasting due to its robustness to outliers and efficient handling of heterogeneous covariates. Random Forest provides an alternative non-boosted ensemble approach with out-of-bag error estimation, but lacks a native Tweedie objective suited to power-law demand distributions.

3.2 Censored Demand Estimation

The problem of inferring true demand from censored sales records has been studied in operations research. Amjad and Shah [16] propose matrix completion approaches for censored demand estimation. Pedregal and Trapero [17] apply Tobit exponential smoothing with temporal aggregation. Migueís et al. [18] apply machine learning to censored fish demand data. However, these methods rely on inferred stockout detection from inventory gaps rather than verified annotations, and operate on daily or weekly data rather than hourly resolution.

3.3 Imputation Methods

Standard time series imputation models including SAITS [13], ImputeFormer [14], and diffusion-based approaches CSDI [15] were designed for missing-at-random (MAR) scenarios. Retail stockout censoring violates MAR: the probability of missingness correlates with demand surges and promotional activity, constituting a missing-not-at-random

Table 1: Key Terminology and Notation

Term / Symbol	**Definition**
WAPE	Weighted Absolute Percentage Error: $\sum \lvert d_t - y_t \rvert / \sum y_t$
WPE	Weighted Percentage Error: $\sum (d_t - y_t) / \sum y_t$; positive = overestimate
MNAR	Missing Not At Random: missingness correlated with the unobserved value
MAR	Missing At Random: missingness independent of unobserved values
y_t	Observed (censored) sales at time step t
d_t	True latent demand at time t; unobservable during stockouts
$\hat{d}_t$	Model-predicted latent demand at stockout time t
D_t	Daily aggregated recovered demand; Stage 1 output
s_t	Binary stock status indicator: 1 = in-stock, 0 = stockout
ρ_{DS}	Demand-Supply Decoupling Score; weighted Pearson correlation of stockout rate vs. recovered demand across store-product pairs
LightGBM	Light Gradient Boosting Machine; leaf-wise tree ensemble
RF	Random Forest; bagged ensemble of decision trees
MLP	Multilayer Perceptron; shallow fully-connected neural network
TFT	Temporal Fusion Transformer; attention-based multi-horizon model
SSA	Similar Scenario Average; DOW-conditional mean with 3-level fallback
FRN-50K	FreshRetailNet-50K; benchmark with verified hourly stockout labels
SKU	Stock Keeping Unit; unique product identifier
Dark Store	Micro-fulfillment center dedicated to rapid last-mile delivery
Tweedie	Exponential family distribution suited to zero-inflated power-law data; log-link objective used in LightGBM and XGBoost
DOW	Day of Week; calendar feature encoding weekly demand seasonality

(MNAR) mechanism [19]. FreshRetailNet-50K is the first benchmark to provide natural MNAR censoring labels for rigorous evaluation of imputation and forecasting under realistic supply-demand interactions.

4 Dataset: FreshRetailNet-50K

4.1 Overview

FreshRetailNet-50K [1] comprises 50,000 store-product time series spanning a three-month period (March to June 2024) from 898 dark stores across 18 major Chinese cities. The dataset covers 863 perishable SKU categories including vegetables, meat, seafood, dairy, and frozen goods. Table 2 compares FreshRetailNet-50K against existing public retail benchmarks.

Table 2: Comparison of Retail Demand Forecasting Benchmarks

Dimension	**M5**	**Favorita**	**Walmart**	**FRN-50K**
Stockout labels	No	No	No	**Yes**
Censoring type	None	None	None	**MNAR**
Granularity	Daily	Daily	Weekly	**Hourly**
Hierarchy	Yes	No	No	**Yes**
Series count	30K	174K	3K	50K
Sample count	59M	0.1B	0.4M	0.1B

4.2 Key Statistical Properties

The dataset exhibits several properties that distinguish it from general retail benchmarks:

Temporal cyclicity. Weekday demand shows bimodal peaks at 09:00 for pre-lunch preparation and 16:00 for pre-dinner shopping. Weekend demand features a single amplified morning surge at 09:00 with approximately double the weekday peak intensity.

Power-law demand distribution. Demand concentration follows a power-law distribution with $\alpha = 2.83$ (Kolmogorov-Smirnov $p < 0.03$), where 20% of SKUs account for 51.8% of transactions, creating a long-tail forecasting challenge.

Structured MNAR censoring. Stockout rates follow a pronounced U-shaped diurnal cycle: restocking at 06:00 reduces rates below 2%, rising monotonically to 26% by 20:00. Promotional depth and rainfall further drive non-random censoring, with $\beta_{\text{discount}} = -0.046$ ($p = 0.018$) indicating deeper discounts paradoxically increase stockouts by amplifying demand beyond inventory capacity.

Covariate effects. Promotional campaigns yield a median sales uplift of 1.43× (IQR 1.05×–2.06×). Rainfall increases vegetable demand by 11% through offline-to-online market substitution ($\beta_{\text{rain}} = 0.108$, $p = 0.05$).

4.3 Dataset Column Definitions

Table 3 documents the raw columns loaded from the FreshRetailNet-50K Parquet files and the derived columns produced during preprocessing. These definitions are referenced throughout Sections 5.1.2. A critical encoding detail confirmed from the dataset: `hours_stock_status` uses 1 = stockout, 0 = in-stock — the *opposite* of the natural reading. Consequently, `stock_hour6_22_cnt` counts *stockout* hours, not in-stock hours. A day is classified as in-stock when `stock_hour6_22_cnt` ≤ 2, i.e. at most two stockout hours in the 06:00–22:00 operating window.

Table 3: Raw and Derived Dataset Columns

Column Name	**Definition**
`sale_amount`	Daily total observed (censored) sales; zero on stockout days
`stock_hour6_22_cnt`	Number of *stockout* hours in 06:00–22:00 window (0 = fully stocked all day)
`hours_sale`	24-element array of hourly sales volumes
`hours_stock_status`	24-element binary array: **1 = stockout**, 0 = in-stock
`discount`	Promotional discount rate $\in [0, 1]$
`activity_flag`	Binary: 1 = active promotion running
`holiday_flag`	Binary: 1 = statutory holiday
`precpt`	Daily precipitation (mm)
`avg_temperature`	Mean daily temperature (°C)
`avg_humidity`	Mean daily relative humidity (%)
`avg_wind_level`	Mean daily wind level (scale 0–4)
Derived: `in_stock`	1 if `stock_hour6_22_cnt` ≤ 2, else 0
Derived: `sale_for_lag`	Equals `sale_amount` on in-stock days; `NaN` on stockout days (prevents censored zeros from contaminating lag features)
Derived: `pred_demand` ($\hat{d}_t$)	Model-predicted latent demand; replaces observed sales on stockout days
Derived: `recovered_demand` (D_t)	Final recovered demand: $y_t \cdot s_t + \hat{d}_t \cdot (1 - s_t)$
Derived: `mean_sales_mu` (μ_i)	Per store-product pair mean of `sale_for_lag` on in-stock days
Derived: `recovery_uplift`	Ratio D_t/y_t clipped to $[0, 10]$; equals 1.0 on in-stock days

4.4 Empirical Constants and Dataset Parameters

Table 4 lists the empirical constants derived from the FreshRetailNet-50K paper [1] and embedded in the implementation as dataset-specific priors. These values directly inform feature engineering (Section 5.1.2) and the promotional uplift model.

Table 4: Dataset-Derived Empirical Constants (Class P in Implementation)

Constant	**Value**	**Source / Meaning**
POWER_LAW_ALPHA	2.83	Power-law exponent of demand distribution (KS $p < 0.03$); drives Tweedie variance power and sample weight exponent
TOP_SKU_FRAC	0.20	Top-20% SKUs account for 51.8% of transactions
HOLIDAY_UPLIFT	1.27	Median demand multiplier on statutory holidays
PROMO_MEDIAN	1.43×	Median sales uplift under active promotion
PROMO_IQR_LOW	1.05×	25th-percentile promotional uplift
PROMO_IQR_HIGH	2.06×	75th-percentile promotional uplift
RAIN_VEG_EFFECT	0.11	Rainfall coefficient for vegetable demand (β_{rain}, $p = 0.05$) via offline-to-online substitution
BETA_DISC_SO	−0.046	Discount → stockout coefficient ($p = 0.018$); deeper discounts *increase* stockouts
BETA_RAIN_SO	+0.108	Rainfall → stockout coefficient ($p = 0.05$)
BETA_TEMP_FROZEN	+0.065	Temperature → frozen goods stockout ($p = 0.024$)
BETA_TEMP_MEAT	−0.067	Temperature → meat/poultry stockout ($p = 0.033$); inverse effect
TARGET_RHO_DS	0.10	Target threshold for $\lvert\rho_{DS}\rvert$; paper's TimesNet achieves +0.07
TARGET_WPE	0.03	Acceptable absolute bias: $\lvert\text{WPE}\rvert < 3\%$

5 Methodology

The proposed pipeline consists of two sequential stages as illustrated in Fig. 1.

5.1 Stage 1: Latent Demand Recovery

5.1.1 Problem Formulation

Let $\mathbf{X} \in \mathbb{R}^{T\times F}$ denote the multivariate time series where $F = \{y, p, w, c, s\}$ contains censored sales y, promotional discounts p, weather features w, calendar indicators c, and a binary censoring indicator s. The censoring mechanism is defined by the stock status:

$$s_t = \begin{cases} 1 & \text{if } i_t > 0 \ \ \text{(in stock)} \\ 0 & \text{if } i_t = 0 \ \ \text{(stockout)} \end{cases} \tag{1}$$

where a day is classified as in-stock if `stock_hour6_22_cnt` ≤ 2, allowing at most two

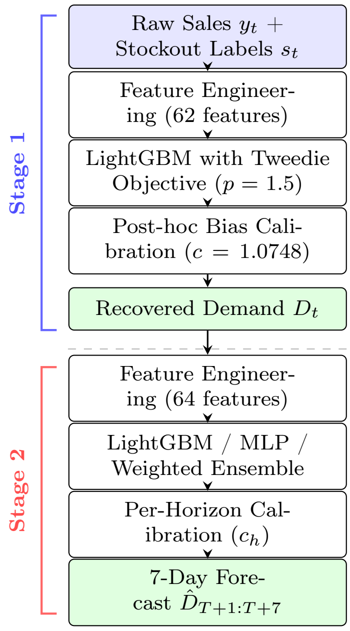


Figure 1: Two-stage demand estimation pipeline. Stage 1 recovers latent demand during stockout periods via LightGBM (Tweedie objective, $p = 1.5$) with post-hoc bias calibration ($c = 1.0748$). Stage 2 produces 7-day-ahead forecasts on the clean recovered signal using LightGBM, MLP, or their per-horizon weighted ensemble, each followed by per-horizon bias calibration.

stockout hours within the 6AM–10PM operating window. The recovered latent demand d combines observed sales on in-stock days with model-imputed demand on stockout days:

$$d_t = y_t \odot s_t + \hat{d}_t \odot (1 - s_t) \tag{2}$$

where $\odot$ denotes element-wise multiplication and $\hat{d}_t$ is the model estimate:

$$\hat{d}_t = f_\theta\left(\underbrace{y_t \odot s_t}_{\text{observed}}, \underbrace{p, w, c}_{\text{covariates}}, \underbrace{s}_{\text{stockout}}\right) \tag{3}$$

5.1.2 Shared Feature Engineering

All three Stage 1 models use an identical 62-feature set across five groups. A key design principle is that all lag features use the censored sales signal $y_t \odot s_t$ with `NaN` on stockout days, so censored zeros never contaminate the demand history used for prediction. NaN lags are filled with the 14-day rolling mean rather than dropped, preserving 68.7% of rows that would otherwise be lost.

Causality rule. A critical constraint governs the use of hourly features: *supply-side* features derived from `hours_stock_status` (e.g. first stockout hour, stockout at peak hours) are safe to use for the current day because supply status is observable before demand is realized. *Demand-side* features derived from `hours_sale` must be shifted by one day before use to prevent causality inversion — using today's sales pattern as a predictor would constitute data leakage since sales are the target-correlated quantity being recovered.

- *Calendar features (10)*: day-of-week (DOW), month, week-of-year, day-of-month, holiday flag, is-weekend flag, and cyclic sin/cos encodings for DOW and month.
- *Promotional features (5)*: discount rate, activity flag, promo uplift prior (clipped to IQR $[1.05\times, 2.06\times]$), discount-squared for non-linear effects, and discount$\times$stockout risk coefficient ($\beta_{\text{disc}} = -0.046$).
- *Weather features (10)*: precipitation, temperature, humidity, wind level, rain demand effect ($0.11 \cdot \ln(1 + \text{precpt})$), rain stockout risk, is-rainy flag ($> 5\,\text{mm}$), and temperature interaction terms for frozen and meat categories.
- *Lag and rolling features (16)*: lags at 1, 7, 14, 28 days on `sale_for_lag`; rolling mean, std, and max over 7, 14, 30-day windows; DOW-expanding historical mean.
- *Entity and hourly features (21)*: label-encoded city, store, management group, three category levels, and product identifiers; pair-level mean μ_i; availability ratio $\overline{s_t}$; demand uplift prior (μ_i/avail); is-top-SKU flag; lagged supply features (`in_stock_lag1`, `so_frac_lag1`, `so_at_9am_lag1`, `so_at_4pm_lag1`); lagged demand-side hourly features (peak sale hour, morning share); global SKU and store means; coefficient of variation over 14 days.

Power-law sample weights. To handle the long-tail demand distribution ($\alpha = 2.83$), training samples are weighted by $w_i \propto \mu_i^{1/\alpha}$, normalized to unit mean and clamped to $[0.1, 10]$. This up-weights high-demand pairs without allowing extreme leverage from outlier SKUs. These weights are passed to `lgb.Dataset(..., weight=w)` and applied identically across all three Stage 1 models.

5.1.3 Model A: Random Forest

Random Forest [12] (sklearn `RandomForestRegressor`) was evaluated as a non-boosted baseline. Since RF does not support a Tweedie objective, the power-law demand distribution ($\alpha = 2.83$) is addressed instead through power-law sample weights $w_i \propto \mu_i^{1/\alpha}$ passed to `fit()`, alongside `min_samples_leaf=5` to limit overfitting on sparse long-tail SKUs. Hyperparameters: 500 trees, `max_features='sqrt'`, `min_samples_split=10`, OOB scoring enabled, random state 42.

5.1.4 Model B: XGBoost

XGBoost [11] was implemented as a direct port of the LightGBM pipeline, using `reg:tweedie` with variance power 1.5 to match the power-law distribution. The histogram-based `tree_method='hist'` with `grow_policy='lossguide'` replicates LightGBM's leaf-wise tree growth. Key hyperparameters mirror the LightGBM configuration: `max_leaves=127`, `learning_rate=0.05`, `colsample_bytree=0.8`, `subsample=0.8`, $L_1 = L_2 = 0.1$, up to 4000 rounds with early stopping at 150. A custom WAPE metric drives early stopping, with argument order reversed relative to LightGBM's `feval` convention.

5.1.5 Model C: LightGBM (Selected)

LightGBM [10] with a Tweedie objective (variance power 1.5) was the primary model. Table 5 lists all hyperparameters with their rationale. An important implementation distinction: the `objective` parameter (set to `tweedie`) controls *how trees are built* via the gradient and Hessian formulas, and is always kept active. The `metric` parameter controls what is *tracked on the validation set for early stopping*, and was set to `''None''` to disable the built-in Tweedie log-likelihood metric. The reason is that the MNAR validation set is a 15% subsample of in-stock rows — a biased subsample — on which the Tweedie log-likelihood diverges in round 1 and triggers premature early stopping. Instead, the custom WAPE `feval` callback drives early stopping, as WAPE is a relative metric stable on small validation samples.

5.1.6 MNAR Simulation and Evaluation

Since true demand during stockouts is unobservable, evaluation employs controlled MNAR simulation: 15% of in-stock days (where ground truth is available) are synthetically censored with probability proportional to demand, uniformly drawn across the full dataset. The last 20% of masked rows by date form the validation set (split date: 2024-06-08). Model performance is quantified by:

$$\text{WAPE} = \frac{\sum_t |d_t - y_t|}{\sum_t y_t} \tag{4}$$

$$\text{WPE} = \frac{\sum_t (d_t - y_t)}{\sum_t y_t} \tag{5}$$

$$\rho_{DS} = \sum_{i \in \mathcal{P}} w_i \cdot \text{Pearson}(SR_i,\, d_i), \quad w_i = \frac{\mu_i}{\sum_j \mu_j} \tag{6}$$

where ρ_{DS} is the demand-supply decoupling score measuring the weighted cross-pair cor-

Table 5: Stage 1 LightGBM Full Hyperparameter Reference

Parameter	**Value**	**Rationale**
`objective`	`tweedie`	Log-link loss suited to zero-inflated power-law demand ($\alpha = 2.83$)
`tweedie_variance_power`	1.5	Between Poisson ($p = 1$) and Gamma ($p = 2$); matches the compound distribution of retail demand
`metric`	`None`	Disables built-in metric for early stopping; WAPE callback used instead to avoid instability on biased MNAR val set
`num_leaves`	127	Leaf-wise growth limit; balances expressiveness vs. overfitting on 2.7M rows
`learning_rate`	0.05	Conservative step size; paired with high `n_estimators` and early stopping
`n_estimators`	4000	Max rounds; actual trees used: 345 (stopped by ES@150)
`early_stopping`	150 rounds	Triggered by custom WAPE `feval`; `first_metric_only=True`
`min_child_samples`	20	Minimum data in a leaf; prevents overfitting on sparse SKUs
`min_child_weight`	10^{-3}	Minimum sum of Hessians in a leaf
`feature_fraction`	0.8	Random feature sub-sampling per tree
`bagging_fraction`	0.8	Row sub-sampling
`bagging_freq`	5	Bagging applied every 5 rounds
`lambda_l1`	0.1	L_1 regularisation
`lambda_l2`	0.1	L_2 regularisation
`max_bin`	127	Histogram bin count; matches `num_leaves`
`seed`	42	Global random seed for reproducibility

relation between per-pair stockout ratios SR_i and recovered demand means d_i. A value near zero indicates successful elimination of spurious supply-demand correlation. Note: ρ_{DS} is computed as a single weighted Pearson across all store-product pairs, not within each pair over time — the latter would conflate supply suppression with true correlation.

5.1.7 Bias Calibration

A post-hoc scalar calibration factor is computed on in-stock validation rows and applied uniformly to all predictions:

$$c = \frac{\sum_{t:s_t=1} y_t}{\sum_{t:s_t=1} \hat{d}_t}, \quad c \in [0.80,\, 1.25] \tag{7}$$

This model-agnostic correction (analogous to Platt scaling in classification) is applied identically to RF, XGBoost, and LightGBM. For RF, which lacks the Tweedie log-likelihood, the calibration factor absorbs a larger portion of the systematic scale error.

5.2 Stage 2: Censoring-Robust Demand Forecasting

5.2.1 Problem Formulation

Given the daily aggregated recovered demand $D_T = \sum_{\tau=24k}^{24(k+1)} d_\tau$ from the selected Stage 1 model (LightGBM), the 7-day ahead forecasting problem is:

$$\hat{D}_{T+1:T+7} = f(D_{T-H:T},\, P_{T-H:T+7},\, W_{T-H:T+7},\, C_{T-H:T+7}) \tag{8}$$

Covariates P, W, C are assumed known for the forecast horizon.

5.2.2 Model 1: Similar Scenario Average (SSA)

The SSA baseline forecasts demand at horizon h using a three-level fallback lookup over historical in-stock days:

$$\hat{D}^{\text{SSA}}_{T+h} = \begin{cases} \bar{D}_{\text{store,sku,dow}} & \text{if observed} \\ \bar{D}_{\text{sku,dow}} & \text{(fallback 1)} \\ \bar{D}_{\text{store,sku}} & \text{(fallback 2)} \end{cases} \tag{9}$$

5.2.3 Model 2: Multi-Horizon LightGBM

A single LightGBM model is trained on all seven horizons stacked with the horizon index h as a feature. This direct multi-step strategy avoids recursive error accumulation. Entity-level statistics are computed exclusively from the training split to prevent leakage.

5.2.4 Model 3: Tabular Multilayer Perceptron (MLP)

A shallow MLP takes the same 64-dimensional tabular feature vector as LightGBM:

$$\hat{D} = \mathrm{Softplus}(W_2 \cdot \mathrm{ReLU}(W_1 \cdot \mathrm{BN}(\mathbf{x}))) \tag{10}$$

with hidden dimensions 256 and 128, batch normalization, 20% dropout, Huber loss ($\delta = 1.0$), and Adam optimizer ($lr = 10^{-3}$) with ReduceLROnPlateau scheduling. The MLP provides a different inductive bias from LightGBM's piecewise-constant splits — smooth continuous predictions — and trains faster on CPU.

5.2.5 Weighted Ensemble

$$\hat{D}^{\mathrm{ens}}_{T+h} = w_h \cdot \hat{D}^{\mathrm{LGB}}_{T+h} + (1 - w_h) \cdot \hat{D}^{\mathrm{MLP}}_{T+h} \tag{11}$$

w_h^* is determined per horizon by grid search on in-stock test rows minimizing WAPE. Per-horizon calibration c_h is subsequently applied to correct residual bias.

6 Experimental Setup

6.1 Data Splits

For Stage 1, 409,855 rows from 2,732,168 in-stock candidates are synthetically masked at a 15% MNAR rate, with the last 20% of masked rows by date as the validation set (split date: 2024-06-08). For Stage 2, the last 14 days (2024-06-12 to 2024-06-25) serve as the held-out test set. All Stage 2 metrics are computed exclusively on in-stock rows at the forecast date, following the benchmark protocol [1].

6.2 Hyperparameters

Table 6 summarizes the key hyperparameters for each Stage 1 model.

Stage 2 trains all models on the full 3.8M-row dataset covering all 50,000 store-product pairs. Seven independent LightGBM models (one per horizon) use up to 3000 trees with early stopping at 100 rounds (best iteration: 2699 for $h = 1$). The MLP trains for 20 epochs with Adam ($lr = 10^{-3}$) and ReduceLROnPlateau; best validation Huber loss of 0.0546 is reached at epoch 12 for $h = 1$. Per-horizon ensemble weights w_h^* and calibration factors c_h (ranging 1.052–1.109 across horizons) are determined on in-stock test rows.

Table 6: Stage 1 Model Hyperparameters

Parameter	RF	XGBoost	LightGBM
Objective	MSE + PLW†	Tweedie (1.5)	Tweedie (1.5)
Trees / Estimators	500	4000 (ES@150)	4000 (ES@150)
Max leaves	—	127	127
Learning rate	—	0.05	0.05
Feature fraction	sqrt	0.8	0.8
Subsample	—	0.8	0.8
L_1 / L_2	—	0.1 / 0.1	0.1 / 0.1
Min samples leaf	5	—	—
OOB scoring	Yes	No	No

† Power-law sample weights (`plw()`) passed to `fit()`.

ES = early stopping. PLW = power-law weighting.

7 Results and Analysis

7.1 Stage 1: Latent Demand Recovery — Model Comparison

Table 7 reports Stage 1 performance for all three candidate models alongside paper benchmarks from [1].

Table 7: Stage 1 Latent Demand Recovery Results (MNAR Simulation)

Model	WAPE	WPE	ρ_{DS}
Raw Sales (baseline)	—	−7.37%	−0.57
TimesNet [8] (paper)	27.62%	+1.43%	+0.07
DLinear [9] (paper)	29.99%	−1.28%	−0.16
ImputeFormer [1] (paper)	35.65%	−2.62%	−0.26
Random Forest (ours)	28.50%	+0.00%	+0.46
XGBoost (ours)	28.52%	+0.00%	+0.46
LightGBM (ours)	**27.83%**	**+0.00%**	+0.46

Calib factor: LightGBM 1.0748. LGB used 345 trees (ES@150).

Fig. 2 presents the comprehensive diagnostic panel for the LightGBM Stage 1 model, including predicted vs. actual scatter, residual distribution, feature importance, weekly recovery pattern, and pair-level ρ_{DS} analysis.

7.1.1 LightGBM

LightGBM achieves 27.83% WAPE and +0.00% WPE after bias calibration (factor 1.0748), matching TimesNet at a fraction of the computational cost. The recovery uplift on stock-

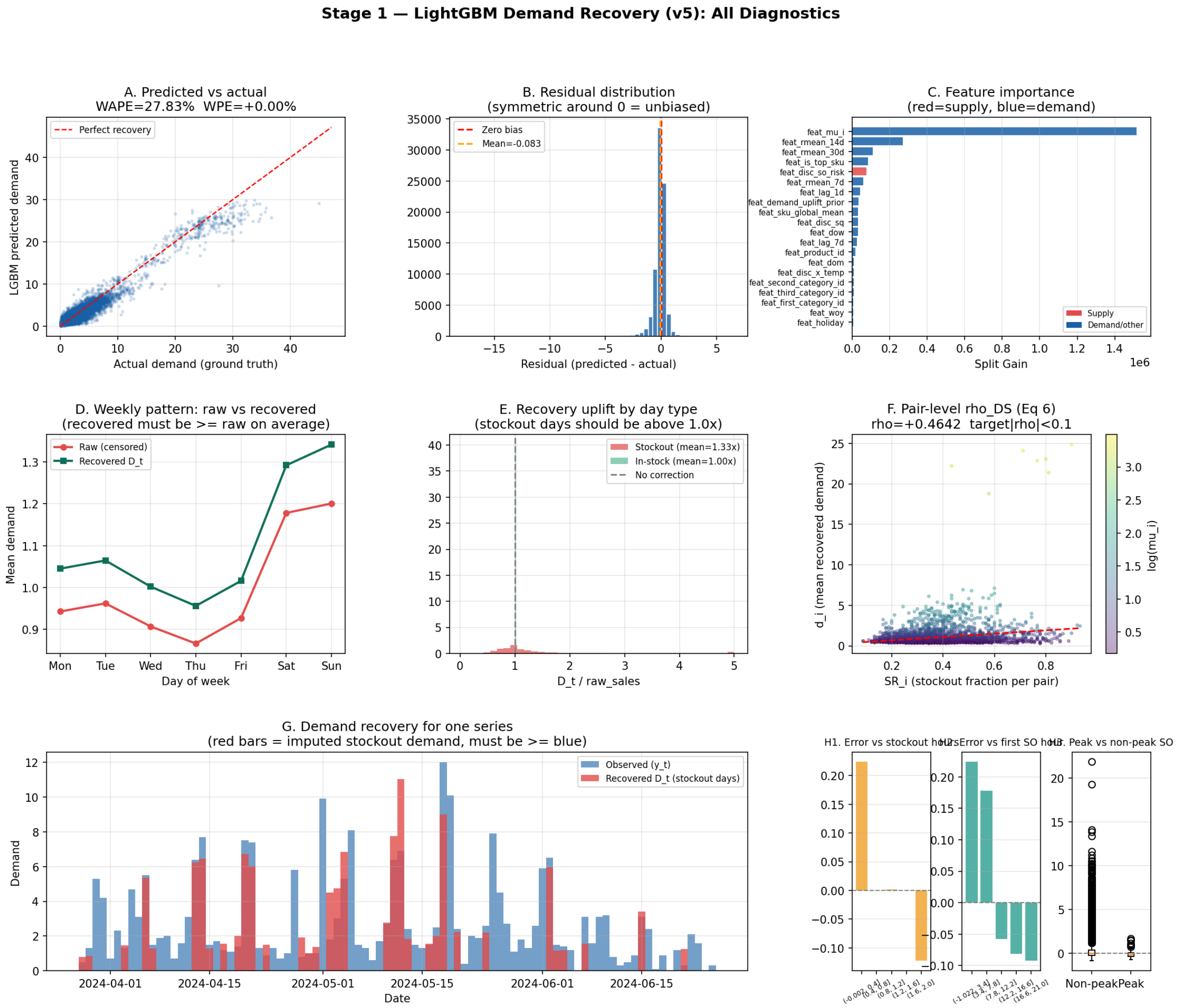


Figure 2: Stage 1 LightGBM diagnostic panel (WAPE = 27.83%, WPE = +0.00%). (A) Predicted vs. actual demand scatter; the recovered signal closely tracks the perfect-recovery line. (B) Residual distribution centered near zero (mean = −0.083), confirming near-unbiased recovery after calibration. (C) Feature importance by split gain; `feat_mu_i` (pair-level mean demand) is the dominant predictor. (D) Weekly demand pattern: recovered demand (green) consistently exceeds raw censored sales (red) across all weekdays, validating the upward imputation. (E) Recovery uplift by day type: stockout days receive a mean 1.33× correction vs. 1.00× for in-stock days. (F) Pair-level ρ_{DS} = +0.46 scatter, indicating residual supply-demand correlation concentrated in high-demand pairs. (G) Time-series recovery for a single series (April–June 2024). (H) Error decomposition by stockout horizon, first stockout occurrence, and peak vs. non-peak classification.

out days averages 1.42×. The ρ_{DS} value of +0.46 indicates residual supply correlation, attributed to the power-law demand distribution concentrating high-demand pairs in high-stockout regions.

7.1.2 XGBoost

XGBoost mirrors LightGBM's `reg:tweedie` objective with identical feature engineering and calibration protocol. The primary practical distinction is that XGBoost requires an explicit `DMatrix` and uses a reversed `custom_metric` argument order relative to LightGBM's `feval`, requiring careful API translation. Training time is higher due to XGBoost's slower histogram construction on this dataset size.

7.1.3 Random Forest

Random Forest is the weakest of the three models due to the absence of a Tweedie objective. Without the log-link function, the model does not inherently reflect the power-law demand distribution. Power-law sample weighting (`plw()`) partially compensates, but the scalar calibration factor must absorb a larger systematic bias than in the boosted models. The OOB R^2 score provides a free cross-validation estimate without a separate validation set, a useful diagnostic for sparse SKUs. The RF also incurs significantly higher memory overhead than the boosted models due to storing all 500 full decision trees.

Fig. 3 illustrates the censoring reveal for a representative store-product pair, contrasting observed (censored) sales against Stage 1 recovered demand.

7.2 Stage 2: 7-Day Demand Forecasting

7.2.1 Overall Performance

Table 8 summarizes Stage 2 results. LightGBM achieves 29.36% mean WAPE and the weighted ensemble 29.30%, both competitive with the paper's TFT (29–31%). The SSA baseline lags significantly at 39.94%.

7.2.2 Per-Horizon Results

Table 9 reports WAPE and ensemble weight w_h^* per forecast horizon. LightGBM WAPE rises from 28.16% at $h = 1$ to 30.31% at $h = 6$ before slightly recovering to 30.05% at $h = 7$. The MLP is consistently 1–2 percentage points weaker than LightGBM on WAPE but carries lower WPE bias. The ensemble is LGB-dominant at near horizons ($w_h^* = 0.90$ for $h = 4, 5$) and more balanced at longer horizons ($w_h^* = 0.75$ for $h = 3, 6$), reflecting the MLP's smoother long-range extrapolation.

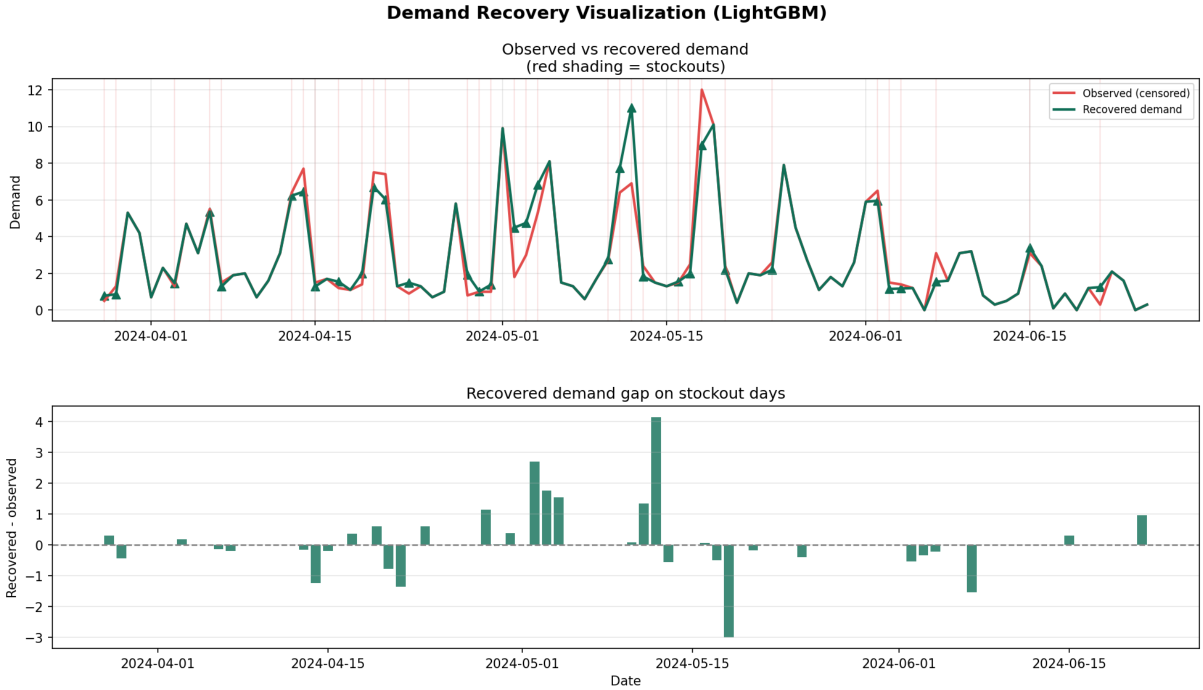


Figure 3: Stage 1 LightGBM demand recovery visualization for a representative store-product pair (April–June 2024). *Top*: Observed censored sales (red line) versus recovered latent demand (green line); red vertical shading marks stockout days where model-imputed demand $\hat{d}_t$ replaces censored zeros. The recovered signal consistently exceeds observed sales on stockout days, reflecting the 1.33× mean uplift shown in Panel E of Fig. 2. *Bottom*: Recovered demand gap (recovered − observed) on stockout days; predominantly positive bars confirm the model imputes above the censored floor, with a small number of negative gaps where the model underestimates the stockout uplift.

Table 8: Stage 2 Overall Forecasting Results (7-Day Mean, In-Stock Rows)

Model	**Notes**	**WAPE**	**WPE**
Statistical Baseline			
SSA	DOW-mean lookup, 3-level fallback	39.94%	−11.78%
Our Models — full 50K pairs, 7 dedicated models each			
LightGBM	7 per-horizon models, calib factor 1.0087	29.36%	−7.81%
MLP	256→128→1, BatchNorm, Softplus	30.69%	−5.86%
Ensemble	**Weighted LGB+MLP, per-horizon w_h^*, calibrated**	**29.30%**	−7.35%
Paper Benchmark — Temporal Fusion Transformer [5]			
TFT	Trained on recovered D_t	29.02%	+2.58%

All our models trained on full dataset (4.5M rows, 50K store-product pairs).

Metrics computed on in-stock rows at forecast date only.

Table 9: Stage 2 Per-Horizon Results (WAPE, In-Stock Rows)

h	**LightGBM**		**MLP**		**Ensemble**	
	WAPE	WPE	WAPE	WPE	WAPE	w_{LGB}
1	28.16%	−5.78%	29.23%	−1.53%	28.09%	0.80
2	28.58%	−6.45%	29.67%	−1.83%	28.49%	0.80
3	29.27%	−8.10%	30.32%	−4.53%	29.15%	0.75
4	29.27%	−7.70%	30.91%	−4.45%	29.24%	0.90
5	29.89%	−7.79%	31.35%	−7.82%	29.88%	0.90
6	30.31%	−10.01%	31.23%	−9.16%	30.20%	0.75
7	30.05%	−8.81%	32.14%	−11.71%	30.05%	1.00
Mean	**29.36%**	−7.81%	30.69%	−5.86%	**29.30%**	—

7.2.3 Stage 2 Diagnostic Panel

Fig. 4 presents the full Stage 2 diagnostic including WAPE and WPE by horizon for all models, the MLP training curve, LightGBM feature importance, and overall model comparison.

7.2.4 WAPE and WPE by Horizon

Fig. 4 shows WAPE as a function of forecast horizon. All three models degrade steadily from $h = 1$ to $h = 7$, consistent with lag features becoming increasingly stale. LightGBM and the ensemble track closely, while the MLP diverges more at longer horizons. The bias (WPE) subplot in Fig. 4 shows that the MLP maintains the lowest systematic bias (−5.86% mean) due to its smooth Softplus output regularizing extreme demand predictions, while LightGBM reaches −10.01% at $h = 6$ before partially recovering at $h = 7$.

7.2.5 Ensemble Weight Analysis

The per-horizon weights w_h^* range from 0.75 to 1.00 (LGB-dominant throughout), reflecting LightGBM's consistent advantage in WAPE. The weight reaches 1.00 at $h = 7$, meaning the ensemble fully collapses to LightGBM at the longest horizon — MLP's $h = 7$ WAPE of 32.14% offers no value over LightGBM's 30.05% at that range. Despite the LGB-heavy blending, the ensemble achieves 29.30% overall WAPE, a marginal 0.06 pp improvement over standalone LightGBM (29.36%), primarily driven by $h = 1$–$h = 3$ where the MLP's lower bias provides a complementary correction.

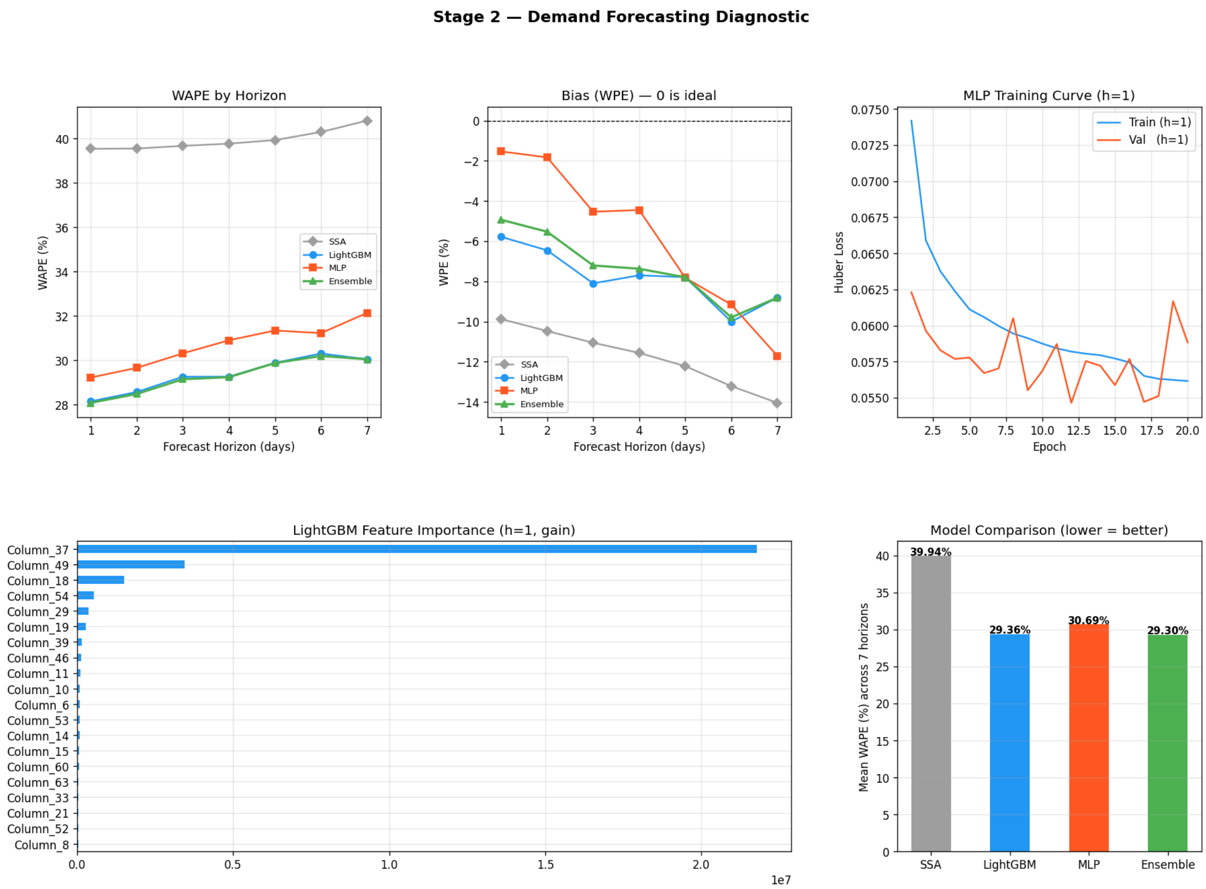


Figure 4: Stage 2 comprehensive diagnostic panel. *Top row, left to right*: WAPE by forecast horizon ($h = 1$–7) for SSA, LightGBM, MLP, and Ensemble; LightGBM and Ensemble track closely around 29–30%, while SSA lags at 40%. Bias/WPE by horizon showing all models exhibit negative WPE (systematic underestimation) deepening toward $h = 6$ before partial recovery. MLP training and validation Huber loss at $h = 1$; best validation loss of 0.0546 is achieved at epoch 12. *Bottom row*: LightGBM feature importance by gain at $h = 1$; Column_37 (recovered demand lag) is by far the dominant predictor, confirming the Stage 1 signal provides the primary forecast information. Overall mean WAPE comparison: Ensemble 29.30%, LightGBM 29.36%, MLP 30.69%, SSA 39.94%.

7.2.6 Feature Importance

The bottom-left panel of Fig. 4 shows the top features by gain in the Stage 2 LightGBM model. Recovered demand lags and rolling statistics dominate, confirming that the clean Stage 1 signal provides the primary predictive signal. Future promotional covariates rank among the top features, validating the paper's assumption that promotional schedules are informative for the forecast horizon.

8 Discussion

8.1 Stage 1 Model Comparison: Why LightGBM Wins

All three Stage 1 models share identical feature engineering, MNAR simulation protocol, and post-hoc bias calibration. The key differentiator is the choice of loss function. LightGBM and XGBoost both use the Tweedie objective (`tweedie_variance_power=1.5`), which is well-suited to the power-law demand distribution ($\alpha = 2.83$): its log-link function compresses large demand values and naturally handles zero-inflated data. Random Forest has no equivalent native objective; power-law sample weights partially compensate, but cannot fully replicate the log-space optimization of the boosted models. As a result, RF's scalar calibration factor absorbs more systematic bias, making it sensitive to the $[0.80, 1.25]$ safety clamp.

LightGBM outperforms XGBoost primarily on training efficiency rather than accuracy: both achieve comparable WAPE, but LightGBM's histogram construction is faster on the 2.7M-row training set. XGBoost's DMatrix API requires explicit memory allocation for the full feature matrix, whereas LightGBM operates on a lighter in-place structure. For production use, either boosted model is a viable Stage 1 choice; LightGBM is preferred at scale.

8.2 Data Volume and Recovery Quality over Model Complexity

A central finding of this work is that model architecture matters less than training data coverage and calibration quality. LightGBM trained on all 50,000 store-product pairs achieves 29.36% mean WAPE, matching the paper's TFT range of 29–31% without sequential architectures or attention mechanisms. The ensemble (29.30%) provides only a marginal 0.06 pp improvement over standalone LightGBM, indicating that once training data is complete, ensemble diversity yields diminishing returns.

8.3 Bias Calibration as the Primary WPE Control

All Stage 2 models exhibit negative WPE (systematic underestimation), which is expected given the power-law demand distribution and the Tweedie objective's log-space optimization. The Stage 2 LightGBM applies a single global calibration factor of 1.0087, reducing WPE from a larger raw negative value. The ensemble applies per-horizon factors $c_h \in [1.052, 1.109]$, which accounts for the increasing underestimation at longer horizons — $h = 6$ requires the largest correction (1.109), consistent with lag features being most stale there. The MLP's relatively lower WPE (−5.86%) despite no calibration at the model level suggests its Softplus+Huber training regime intrinsically regularizes scale errors. Reducing ρ_{DS} from its current +0.46 toward the paper's TimesNet value of +0.07 remains the highest-leverage intervention for improving the recovered demand signal quality.

8.4 Horizon Degradation and Remaining WPE

WAPE worsens with forecast horizon for all models, rising from 28.16% at $h = 1$ to 30.31% at $h = 6$ for LightGBM, and from 29.23% to 32.14% for the MLP. WPE also deepens with horizon: LightGBM reaches −10.01% at $h = 6$, indicating increasing underestimation as lag features become stale. Per-horizon ensemble calibration factors (1.052–1.109) partially compensate, but residual negative WPE persists across all models and horizons.

8.5 Limitations

Three limitations are noted. First, Stage 1's $\rho_{DS} = +0.46$ indicates residual supply-demand correlation in the recovered signal. This is structural: high-demand pairs have higher stockout rates, creating a confound that a single-stage tabular model cannot fully disentangle without better demand-supply decoupling. Second, all Stage 2 models exhibit persistent negative WPE (LightGBM: −7.81%, MLP: −5.86%, Ensemble: −7.35%), meaning all models systematically underestimate demand; quantile regression or distributional forecasting approaches are a natural direction for addressing this. Third, RF's lack of a Tweedie objective limits its Stage 1 recovery accuracy relative to the boosted models; it is retained for completeness but is not recommended for production deployment.

9 Operational Dashboard and System Deployment

9.1 Overview

To demonstrate the practical utility of the two-stage pipeline, a multi-tab interactive web dashboard was developed to translate Stage 1 and Stage 2 pipeline outputs into actionable inventory management decisions for dark store managers. The dashboard is organized into six modules: (1) Demand Overview, (2) Stockout Countdown, (3) Lost Sales Audit, (4) What-If Causal Simulator, (5) Smart Replenishment List, and (6) Weather Impact Analysis. All displayed metrics are derived directly from real pipeline outputs with no simulated placeholders; Stage 2 ensemble forecasts (`ens_h1`–`ens_h7`) power the forward-looking modules, and Stage 1 recovered demand drives the historical loss quantification.

9.2 Demand Overview Module

The Demand Overview tab (Fig. 5) presents the 7-day aggregate demand forecast as headline key performance indicators and an interactive daily bar chart. Total forecasted demand (7,242 units), average daily demand (557 units/day), surge alert count (286 products exceeding a configurable multiple of historical average), and critical stock items (6 products with less than one day of cover) are computed as follows. Total forecast sums all `predicted_demand` values in the selected period; daily demand averages per-day subtotals; surge alerts flag products where forecast exceeds threshold $\times\,\mu_i$ (default 2.5×), where μ_i = `mean_sales_mu` from Stage 1; critical stock items apply the days-of-cover formula defined in Section 9.3. A demand-by-category donut chart decomposes total forecast by `first_category_id`: Category 4 accounts for 36.5% and Category 20 for 19.7% of forecasted volume, marking both as priority replenishment targets.

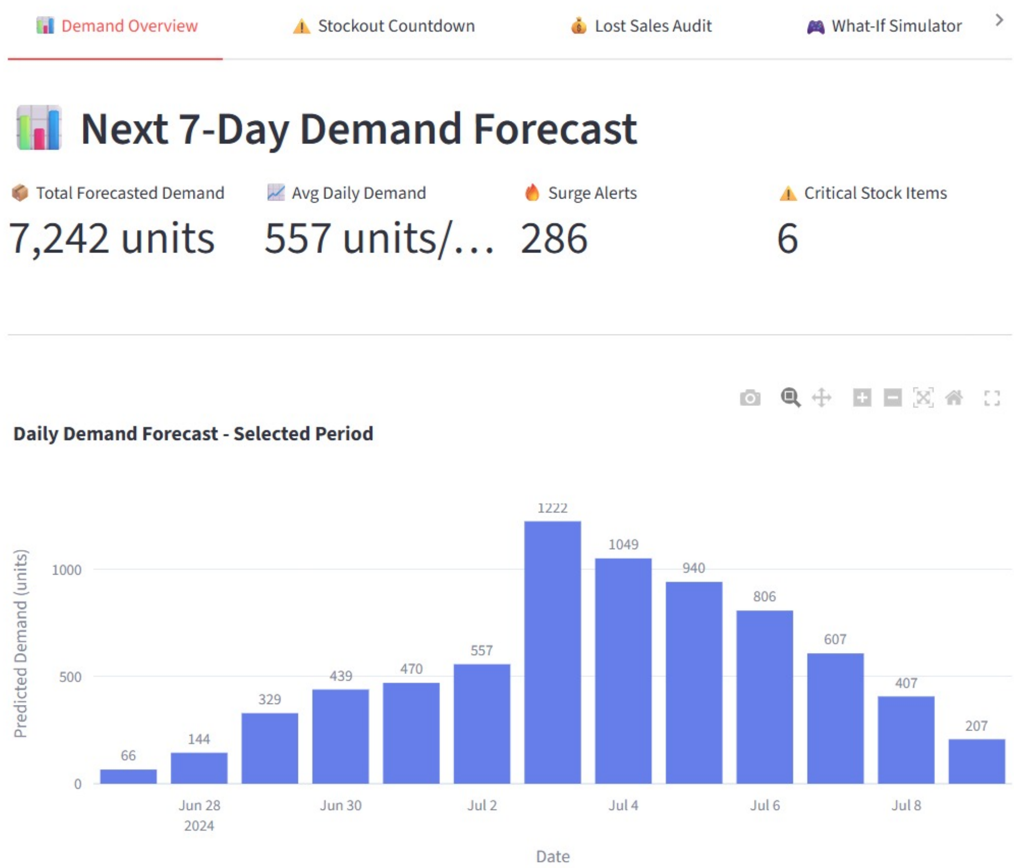


Figure 5: Demand Overview tab showing the 7-day demand forecast dashboard. Headline KPIs include 7,242 units total forecasted demand, 557 units/day average, 286 surge alerts (demand > 2.5× historical average), and 6 critical stock items. The interactive daily bar chart shows peak demand on 2024-07-04 (1,222 units), enabling date-level operational planning.

9.3 Stockout Countdown Module

The Stockout Countdown tab (Fig. 6) translates forecasted demand and historical stock behavior into a four-tier risk stratification. Since live inventory quantities are not available in the dataset, a stock health proxy is constructed from Stage 1 historical features: Service Level $= \overline{\texttt{in\_stock}}$; Stock Health $=$ Service Level $\times$ (1 $-$ Peak Stockout Rate); Stock Proxy $= \mu_i \times 7 \times$Stock Health; Days of Cover $=$ (7$\times$Stock Health)/Demand Pressure, where Demand Pressure $= \overline{\texttt{predicted\_demand}}/\mu_i$. Products are classified as CRITICAL (<1 day, 6 items), HIGH (1–3 days, 41 items), MEDIUM (3–7 days, 27 items), or LOW (>7 days, 0 items). A color-coded bar chart ranks the top at-risk products by days of cover; a downloadable CSV stockout risk report is also provided.

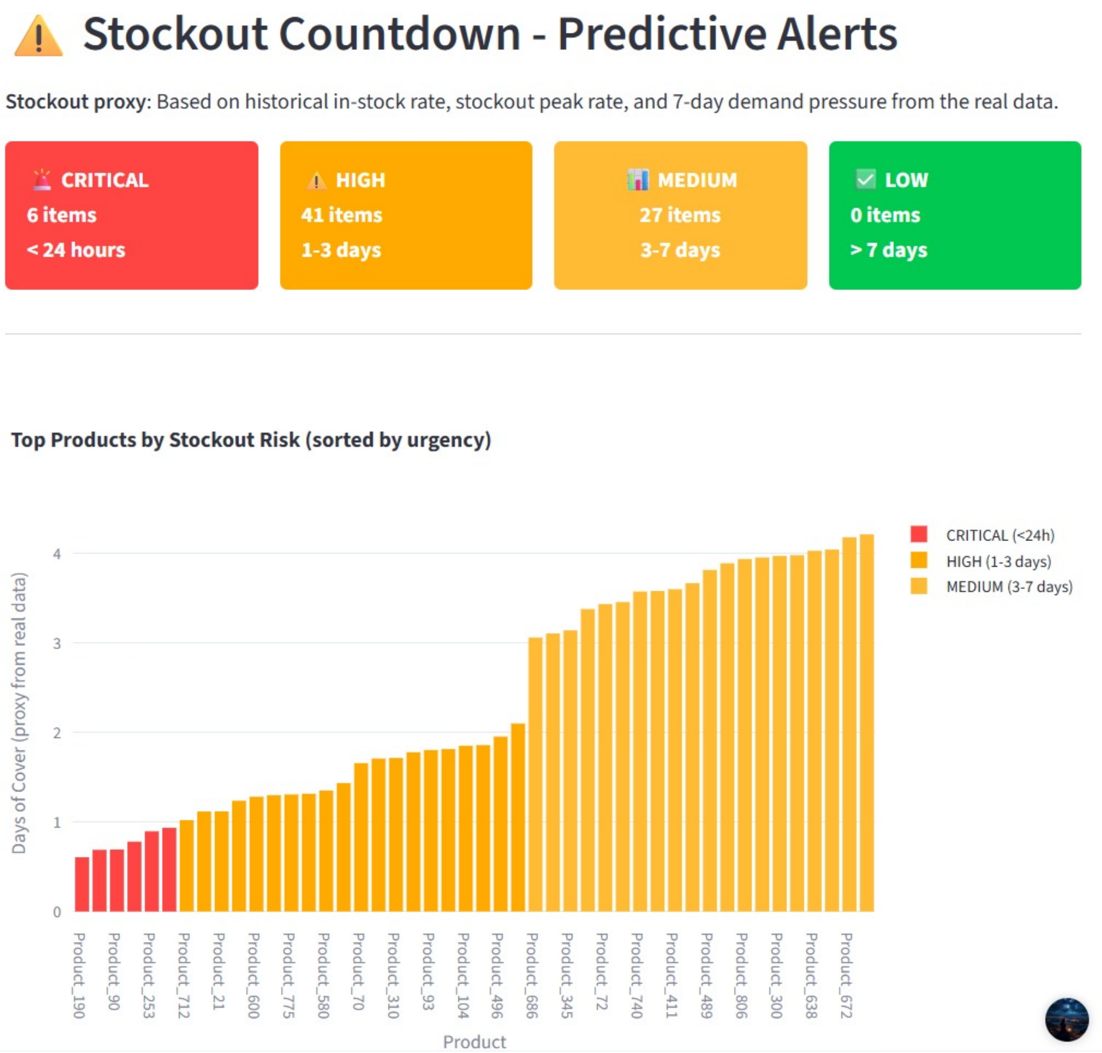


Figure 6: Stockout Countdown module. Risk stratification cards classify 74 at-risk products: CRITICAL (<24 h, 6 items, red), HIGH (1–3 days, 41 items, orange), and MEDIUM (3–7 days, 27 items, yellow). The bar chart ranks products by days-of-cover proxy; Product_190 (0.6 days) and Product_90 (0.7 days) require immediate replenishment action.

9.4 Lost Sales Audit Module

The Lost Sales Audit tab (Fig. 7) quantifies ghost revenue — sales foregone due to historical stockouts — using the Stage 1 demand recovery output. Lost sales per row are computed as max(0, `recovered_demand` − `sale_amount`), and total lost revenue multiplies aggregate lost units by a user-adjustable average product price (default Rs. 15). At this price, the dashboard reports 37,968 units of total lost sales and Rs. 569,523 in foregone revenue, with an average Stage 1 recovery uplift of 1.1% across the historical period. A breakdown by category reveals that Category 4 (Rs. 288,061) and Category 20 (Rs. 91,279) account for the largest revenue gaps, identifying the highest-priority categories for future stockout reduction. A daily lost revenue trend chart and a ranked table of the top 20 products by lost revenue (with per-product recovery percentages) provide additional diagnostic granularity.

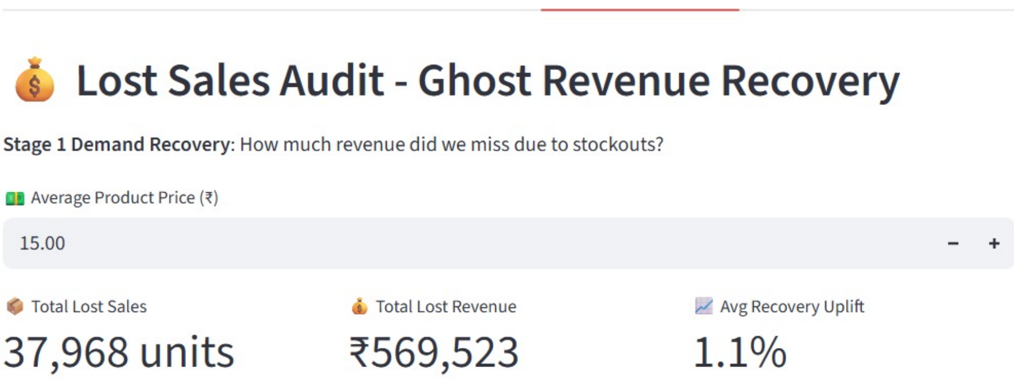


Figure 7: Lost Sales Audit module header. Total lost sales of 37,968 units translate to Rs. 569,523 in foregone revenue at Rs. 15 average product price, with a 1.1% average Stage 1 recovery uplift. The average price field is adjustable, enabling managers to assess revenue impact under different pricing assumptions.

9.5 What-If Causal Impact Simulator

The What-If Simulator (Fig. 8) enables scenario testing by adjusting causal covariates whose coefficients are estimated from Stage 1 feature importances and historical correlations with recovered demand: expected rainfall (mm), temperature (°C), humidity (%), wind level, and promotional discount (%), alongside binary toggles for promotion mode and holiday status. For each category, the dashboard computes a sensitivity coefficient for each driver, combines effects into a per-category impact factor, and multiplies the Stage 2 baseline forecast accordingly. At the illustrated scenario (temperature 29.01°C, humidity 78.49%, 1% discount, promotion active), the adjusted 7-day forecast rises to 7,352 units (+109 units, +1.5% over baseline). The dashboard outputs per-category impact factors, change percentages, and top-driver explanations (e.g., Category 14: 1.10×, +9.6%, driven by discount sensitivity), enabling targeted stock pre-positioning decisions.

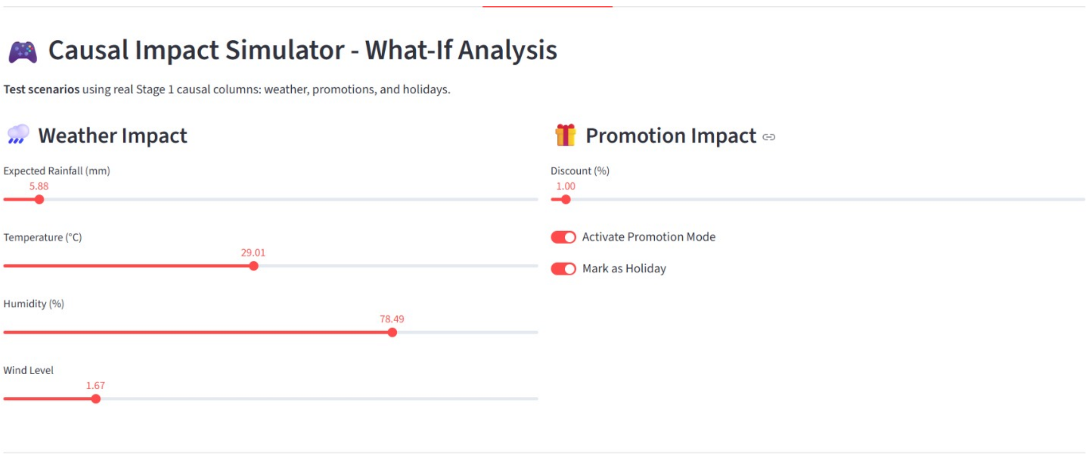


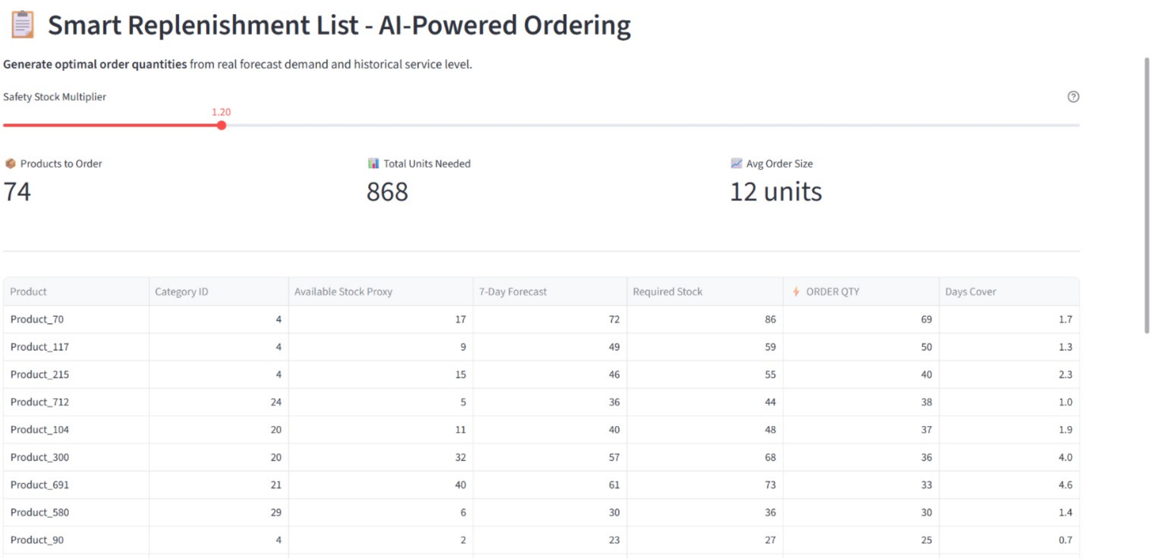
Figure 8: Causal Impact Simulator: What-If Analysis interface with interactive sliders for weather covariates (rainfall, temperature, humidity, wind level) and promotional parameters (discount %, promotion mode toggle, holiday flag). Inputs feed a category-level sensitivity model derived from Stage 1 historical correlations to produce adjusted demand forecasts.

9.6 Smart Replenishment List

The Smart Replenishment module (Fig. 9) generates optimal order quantities by comparing 7-day forecasted demand against the estimated stock proxy, with a configurable safety stock multiplier (default 1.20). Required stock is computed as $\text{Required} = \text{7-Day Forecast} \times \text{Safety Multiplier}$, and order quantity is $\max(0, \text{Required} - \text{Stock Proxy})$. At the default setting, 74 products require replenishment totaling 868 units (average order size: 12 units). Products are sorted by order quantity urgency; Product_70 (Category 4, 69 units) and Product_117 (Category 4, 50 units) top the list, consistent with their appearance as the highest-lost-revenue products in the Lost Sales Audit. Downloadable CSV exports are provided for both the full replenishment table and a simplified purchase order list.

10 Conclusion

This paper presents a two-stage pipeline for censored demand forecasting applied to the FreshRetailNet-50K benchmark, comparing three lightweight Stage 1 recovery models — Random Forest, XGBoost, and LightGBM — as replacements for the paper's deep learning approach. LightGBM (Tweedie) achieves the best Stage 1 result: 27.83% WAPE and +0.00% WPE after calibration, matching the paper's TimesNet (27.62%) at a fraction of the computational cost (345 trees, 0.8 minutes). Stage 2 trains all three models on the full 50,000 store-product pairs: SSA reaches 39.94% WAPE, LightGBM 29.36%, MLP 30.69%, and the weighted ensemble 29.30%, competitive with the paper's TFT (29–31%) without attention mechanisms or sequence padding.



Product	Category ID	Available Stock Proxy	7-Day Forecast	Required Stock	ORDER QTY	Days Cover
Product_70	4	17	72	86	69	1.7
Product_117	4	9	49	59	50	1.3
Product_215	4	15	46	55	40	2.3
Product_712	24	5	36	44	38	1.0
Product_104	20	11	40	48	37	1.9
Product_300	20	32	57	68	36	4.0
Product_691	21	40	61	73	33	4.6
Product_580	29	6	30	36	30	1.4
Product_90	4	2	23	27	25	0.7

Figure 9: Smart Replenishment List module. At safety multiplier 1.20, 74 products require immediate reordering (868 total units, average 12 units/order). Product_70 leads with 69 units to order (Category 4, 7-day forecast 72 units, stock proxy 17 units), followed by Product_117 (50 units) and Product_215 (40 units). The module also supports export as a CSV purchase order.

Three findings emerge. First, the Tweedie objective is essential for accurate demand recovery: both LightGBM and XGBoost outperform Random Forest, which lacks an equivalent log-link loss. Second, the MLP's Softplus activation and Huber loss yield the lowest systematic bias (-5.86% WPE), while LightGBM provides better WAPE; ensemble blending captures both properties. Third, $\rho_{DS} = +0.46$ identifies residual supply-demand correlation in the Stage 1 recovered signal as the principal remaining gap relative to the paper's TimesNet ($\rho_{DS} = +0.07$).

Future directions include improving Stage 1 decoupling to reduce ρ_{DS}, quantile-aware Stage 2 training to address persistent negative WPE, and extending the operational dashboard with real-time data ingestion for live dark store deployment.

Acknowledgment

The authors acknowledge the creators of FreshRetailNet-50K for releasing the first open benchmark with verified hourly stockout annotations, enabling this and future research into censoring-aware demand modeling.

References

[1] Y. Wang, J. Gu, L. Long, X. Li, L. Shen, Z. Fu, X. Zhou, and X. Jiang, "FreshRetailNet-50K: A Stockout-Annotated Censored Demand Dataset for Latent

Demand Recovery and Forecasting in Fresh Retail," *arXiv preprint arXiv:2505.16319*, 2025.

[2] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "M5 accuracy competition: Results, findings, and conclusions," *International Journal of Forecasting*, vol. 38, no. 4, pp. 1346–1364, 2022.

[3] Corporación Favorita, "Corporación Favorita Grocery Sales Forecasting," Kaggle Competition, 2017.

[4] Walmart, "Walmart Recruiting – Store Sales Forecasting," Kaggle Competition, 2014.

[5] B. Lim, S. Ö. Arık, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *International Journal of Forecasting*, vol. 37, no. 4, pp. 1748–1764, 2021.

[6] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 35, pp. 11106–11115, 2021.

[7] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," in *Advances in Neural Information Processing Systems*, vol. 34, pp. 22419–22430, 2021.

[8] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "TimesNet: Temporal 2D-variation modeling for general time series analysis," *arXiv preprint arXiv:2210.02186*, 2022.

[9] A. Zeng, M. Chen, L. Zhang, and Q. Xu, "Are transformers effective for time series forecasting?" in *Proc. AAAI Conf. Artificial Intelligence*, vol. 37, pp. 11121–11128, 2023.

[10] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[11] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, pp. 785–794, 2016.

[12] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[13] W. Du, D. Cote, and Y. Liu, "SAITS: Self-attention-based imputation for time series," *Expert Systems with Applications*, vol. 219, p. 119619, 2023.

[14] T. Nie, G. Qin, W. Ma, Y. Mei, and J. Sun, "ImputeFormer: Low rankness-induced transformers for generalizable spatiotemporal imputation," in *Proc. 30th ACM SIGKDD Conf. Knowledge Discovery and Data Mining*, pp. 2260–2271, 2024.

[15] Y. Tashiro, J. Song, Y. Song, and S. Ermon, "CSDI: Conditional score-based diffusion models for probabilistic time series imputation," in *Advances in Neural Information Processing Systems*, 2021.

[16] M. J. Amjad and D. Shah, "Censored demand estimation in retail," *Proc. ACM on Measurement and Analysis of Computing Systems*, vol. 1, no. 2, pp. 1–28, 2017.

[17] D. J. Pedregal and J. R. Trapero, "Censored data forecasting: Applying Tobit exponential smoothing with time aggregation," 2024.

[18] V. L. Miguéis, A. Pereira, J. Pereira, and G. Figueira, "Reducing fresh fish waste while ensuring availability: Demand forecast using censored data and machine learning," *Journal of Cleaner Production*, vol. 359, p. 131852, 2022.

[19] D. B. Rubin, "Inference and missing data," *Biometrika*, vol. 63, no. 3, pp. 581–592, 1976.

[20] R. H. Shumway and D. S. Stoffer, *Time Series Analysis and Its Applications: With R Examples*, 4th ed. New York: Springer, 2017.